

VAG
Vilnius Aerosol Generator

***USE AND INSTRUCTION
MANUAL***



One Head Unit is Shown

TABLE OF CONTENT

<u>I. DESCRIPTION</u>	<u>3</u>
<u>II. OPERATION</u>	<u>8</u>
2.1. INSTALLATION OF VAG	8
2.2. PREPARATION FOR AEROSOL GENERATION	9
2.3. GENERATION OF AEROSOL	11
2.4. END OF AEROSOL GENERATION	12
<u>III. CALIBRATION</u>	<u>12</u>
<u>APPENDIX 1. EXAMPLES</u>	<u>14</u>

I. DESCRIPTION

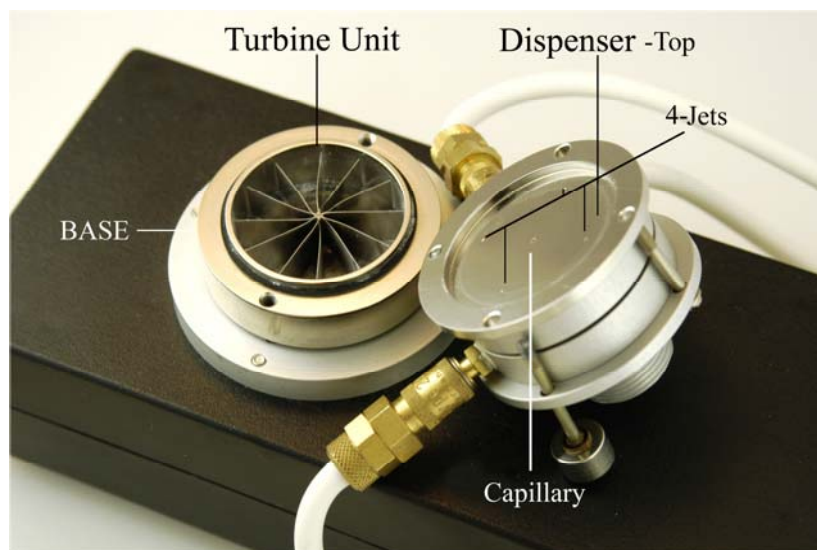
The Vilnius Aerosol Generator (VAG) is a general-purpose dry powder disperser. The VAG is a highly reliable device, which produces aerosol concentrations from 1 to 2000 mg/m³. In addition to its capability of producing high aerosol concentrations, VAG provides exceptional deagglomeration characteristics.



(Two Head Unit is illustrated)

The instrument consists of Controller and Disperser units. The Disperser contains a dynamic aerosolization and a deagglomeration assembly. The aerosolization assembly includes a dust chamber for placement of a powder to be aerosolized. Airflow, directed into the chamber through a control valve and jets impinges upon the powder dispersing it into the air. Simultaneously air jets activate a turbine which is used for continuous supply of powder under the air jets. The aerosolization efficiency is increased by means of vibrating cam which oscillates the turbine.

The aerosolization assembly is interconnected with the deagglomeration assembly by a capillary. The lower end of the capillary is positioned at the top of the dust chamber, while the upper end of the capillary enters the region of high velocity airflow. The aerosol particles are deagglomerated in the deagglomeration assembly by the shear forces that are generated between the high velocity airflow, exiting through a narrow gap around capillary, and the low velocity aerosol stream exiting the capillary.



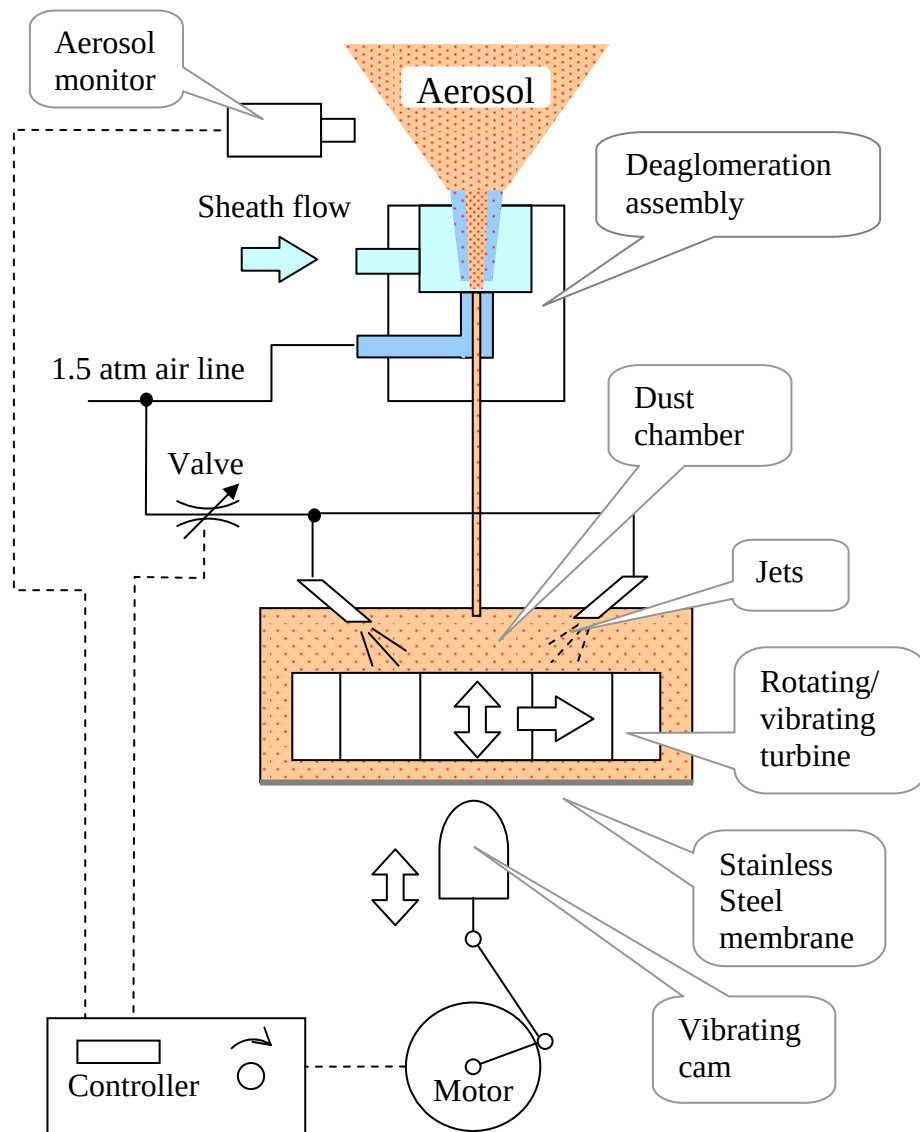


Figure 1. VAG principle of operation

A portable aerosol monitor (Casella MicroDust Pro) is used for continuous monitoring of the aerosol output. The electrical signal from the Casella sensor is used as Controller feed-back signal. If the aerosol output decreases, the Controller causes the turbine to vibrate more actively and increases the air flow rate through the jets. On the contrary, the turbine is slowed down and air flow rate is decreased when aerosol output increases.

The highest stability of aerosol output is achieved when Casella aerosol monitor is employed for monitoring of aerosol output. Other types of real-time aerosol monitors also can be used to monitor aerosol output. However, additional modifications of the electrical circuit must be made for that purpose. Furthermore, the stability of aerosol output may be less condensed and aerosol concentration range reduced if other than the Casella aerosol monitor is used.



The VAG is designed to operate continuously for 0.5 to 6.0 hours depending on the desired aerosol concentration.



II. OPERATION

2.1. INSTALLATION OF VAG



Figure 2. Cable connections

- Plug in Controller.
Make sure that VAG is turned OFF.
- Connect electrical output of Controller to the input of Dispenser.

- Connect three-way cable to the corresponding ports of Casella analog output, VAG Controller MD 880 port and PC COM port.



- Connect Disperser's tubing to the compressed dry air line (1.5 atm pressure).
- Ensure all air connections are tight and secure.

2.2. PREPARATION FOR GENERATION

- Load powder into the dust chamber. For this purpose unscrew two butterfly-screws on Disperser and open the dust chamber. Place powder on the top of turbine. The volume of the loaded powder must not exceed 2/3 of the chamber's volume. Close the chamber. Confirm the seal and O-Ring integrity.



Attach Casella sensor to VAG Dispenser as shown in Figure 3.



Figure 3. VAG Dispenser with Casella sensor mounted on the top.

2.3. GENERATION OF AEROSOL

- Turn Casella aerosol monitor ON.

The aerosol monitor must be prepared and calibrated according to its manual.

- Select appropriate measurement range on the Casella aerosol monitor.



- Select integration constant of 1 s.
- Turn VAG ON.
- Adjust desired aerosol Dispersion Rate. Setting Dispersion Rate, Correction Factor should be taken into account which is estimated during calibration (see chapter III. CALIBRATION).

2.4. END OF AEROSOL GENERATION



- Turn VAG OFF.
- Close compressed air supply line.
- Turn Casella aerosol monitor OFF.
- Clean residual powder from powder reservoir.

III. CALIBRATION

For correct operation the VAG should be calibrated for specific powder type also taking into account VAG application setup. During calibration process the mass of powder dispersed per minute at set concentration level is determined.

The VAG uses the Casella light scattering method for feedback signal generation. This method exhibits some sensitivity to a particulate's refractive index, size, shape, color and density. For correct instrument operation with specific powder type, gravimetric calibration is needed.

The VAG's shelter flow rate is generated by pressure difference between shelter flow injection point and instrument's outlet. Specific application setup like length, diameter and configuration of tubes used for aerosol transportation can influence flow rate and concentration herewith.

Before start of calibration instrument's application, a setup should be defined to take into account its influence on calibration results.

- Weigh powder before loading it into disperser. Prepare VAG and start generation. Set concentration level by turning Controller's knob.
- Let VAG operate until whole powder is dispersed. You'll notice this by rapid decrease of concentration on a graph.
- Note time period and average concentration.
- Calculate Correction Factor from the formula below:

$$\text{Correction Factor} = \frac{\text{Dispersion Rate}}{\text{Instrument Average Reading}} \left(\frac{\text{mg / min}}{\text{mg / m}^3} \right)$$

Note: you can also calibrate an instrument for specified powder type and apply calibration by following the procedures described in Casella user's handbook. Dispersion Rate should be used instead of Gravimetric concentration.

APPENDIX I. CONCENTRATION CHARTS (GLUCOSE POWDER)

